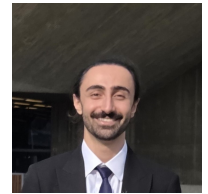


Kasra Mohammadi

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ABOUT ME

I am a Mechanical engineer and MSc Sustainable Energy Technology student at TU Delft, specializing in heating and cooling systems, thermal energy storage, and energy system modelling. I am experienced in developing Python-based engineering models and analyzing ATES (WKO), building energy, and renewable energy systems. Particularly interested in engineering roles involving sustainable heating and cooling, energy infrastructure, and the energy transition.

EDUCATION

Delft University of Technology (TU Delft) M.Sc. Sustainable Energy Technology – Heating and Cooling GPA: 8.9/10	Delft, Netherlands September 2024 – Expected August 2026
University of Tehran (UT) B.Sc. Mechanical Engineering GPA: 19.18/20	Tehran, Iran September 2019 – February 2024

WORK EXPERIENCE

Gemeente Rotterdam – Ingenieursbureau Graduation Project Intern – Thermal Systems and Grid Congestion	Rotterdam, Netherlands January 2026 – Expected August 2026
- Conducting MSc thesis research on grid congestion mitigation using medium-temperature aquifer thermal energy storage (MT-ATES). - Using Python, SEAWAT/MODFLOW, and TESP to assess the potential of MT-ATES to reduce winter peak electricity demand.	
Techniplan Adviseurs BV Intern	Rotterdam, Netherlands September 2025 – December 2025
- Developed a workflow based on SEAWAT, Python, and Excel to simulate subsurface storage conditions of ATES systems. - Made the company's ATES modelling procedure more flexible, automated, and user-friendly.	
Advanced Energy Systems Lab (AESL) – University of Tehran Research Assistant	Tehran, Iran June 2023 – July 2024
- Worked on the design and analysis of a near-zero-energy primary school building with PV integration across different Iranian climates. - The project increased the energy independence of schools from the electricity grid in underdeveloped regions. - I used TRNSYS, EnergyPlus, and MATLAB to obtain, validate, and evaluate the results.	
Department of Geoinformation – AGH University Paid Research Intern	Kraków, Poland Summer 2022
- Performed laboratory tests, processed experimental data, and supported analysis for a wastewater treatment research project. - Assisted in onboarding and guiding another international intern in laboratory procedures and project tasks.	

VOLUNTEERING

Energy for Refugees (EfR) Board Member and Engineer	Delft, Netherlands January 2025 – January 2026
- Designed and sized a self-sustaining PV-battery system for a vocational school in Kakuma refugee camp, Kenya. - Estimated demand profiles, optimized system sizing under funding constraints. - The project resulted in a 58.5 kWp solar installation currently installed and operational.	
Delft Sustainable Energy Association (DelftSEA) Board Member and Treasurer	Delft, Netherlands November 2024 – November 2025
- Managed the association's finances, budgeting, and cash flow to support the association in achieving its goals. - Helped with the organization of student events, improving the social and academic environment.	
Tehran University Emergency Board of Studies Member	Tehran, Iran September 2022 – September 2023
- Member of the Emergency Board of Studies formed during the 2022 national student protests in Iran. - My role as a member was to conduct meetings with professors and academic staff to prevent hindrances to the educational quality of the course and help students express their demands during the protests.	

ACADEMIC EXPERIENCE

Teaching Assistant – TU Delft and University of Tehran

September 2020 – February 2026

- Teaching assistance for courses with a combined study load of over 70 ECTS.
- Responsibilities included tutoring, recitation classes, assignment and quiz design, grading, and guiding students during projects.
- Course in TU Delft: SET3061 – Energy System Modelling
- Courses in University of Tehran: Optimization of Mechanical Systems, Thermodynamics I, Fluid Mechanics I, Physics I, Numerical Calculations

SKILLS

Programming and Modelling: Python, MODFLOW, FloPy, MATLAB, C++, EES

Energy and Simulation Tools: TRNSYS, EnergyPlus, TESPpy, CoolProp, WANDA, COMSOL

Engineering Software: SolidWorks, ANSYS Fluent, MSC ADAMS

Other: Teaching, technical writing, photography

LANGUAGE SKILLS

- Persian: Native
- English: Fluent (C2)
- Turkish: Fluent (C2)
- Dutch: Intermediate (B1)
- Italian: Intermediate (B1)

AWARDS AND HONOURS

- Holder of the “National Examination Top Rank Prize” scholarship awarded by the Fanni Foundation, 2019–2023
- Ranked **46th** among ~160,000 participants in the 2019 National Iranian University Entrance Exam (Mathematics and Physics Field)
- Ranked **5th** in the nation among ~150,000 participants in the 2019 National Iranian University Entrance Exam (Foreign Languages Field)
- Member of Iran’s National Elites Foundation (INEF), 2019 – 2024

REFERENCES

References available upon request.